

Final Year Project Report

Full Unit - Final report

A Concurrency Based Game Environment

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Declaration

This report has been prepared on the basis of my own work. Where other published and unpublished source materials have been used, these have been acknowledged.

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Chapter 1: Abstract

This project is a Real Time Strategy (RTS) Game that focuses on using concurrency to provide a Local Multiplayer (LCM) experience through split screen. This project's primary aim is to provide a comfortable and seamless LCM experience, for both players on controller and mouse. A secondary goal for this project is to provide steady performance regardless of the game state or number of players, aiming to not drop below the users display refresh rate.

Users of this project are able to play with 1-5 people on the same device and display, with 1 player on a keyboard and mouse and up to 4 on different controllers. This project assigns regions of the display to each player and automatically adjusts the aspect ratio and size of pre-existing regions when a player connects. Another feature of this project is seamless disconnection and reconnection for users, a player can disconnect then reconnect their controller to continue playing at any time.

The primary motivation for this project is the poor performance of pre-existing RTS games, even those that focus solely on a single player experience. Other RTS games experience input delay and stuttering issues as the game progresses, primarily due to high unit counts and complex interactions between units. For example, Stellaris by Paradox Interactive is well known by the RTS community to slow down significantly in the late game, to the point that the game cannot be played.

Another Motivation for this project is the lack of split screen multiplayer in the RTS genre. Split screen is often exclusively implemented in First Person Shooter (FPS) and racing Games, although even those genres have seen a decline in split screen as of recently [2]. This project aims not only to revive the once beloved feature of Split Screen but also to introduce it to the RTS genre. RTS is by nature a game with complex interactions between players, I believe the addition of face-to-face communication could overall benefit the experience.

By combining the technical innovations of modern game development with older styles of gameplay featuring more face to face interaction, this project demonstrates that performant, social RTS gaming is both viable and practical.

1.0.1 Aims and Objectives of the project

Primary Objectives

Performance Optimization

- Achieve and maintain a consistent frame rate that initially matches that of the end users maximum refresh rate, and doesn't drop below the minimum video frame rate of 23FPS for any given scenario
- Maintain low input latency that doesn't spike above 50ms in any given scenario but should match that of the end users minimum input latency initially
- Produce no additional input delay or lag spikes regardless of the number of split screen instances running

local multiplayer functionality

- Implement a split screen local multiplayer system that allows for 1-5 players to play the game simultaneously
- Develop a robust input handling system that provides consistent input delay across all game instances

long term stability

- Optimize storage and management of in game units to minimize the performance degradation for a large numbers of units
- Prevent memory leaks to ensure the games memory usage doesn't increase after extended play time, ultimately resulting in a program crash

user experience

- Design a clean UI that is free of clutter and provides a consistent experience across different input devices
- Implement a sensible text size and UI scaling to ensure visibility regardless of how small a given split screen instance is

Secondary Objectives**Comprehensive real time strategy game**

- Survey and respond to real time strategy players preferences
- Implement a progression system that allows players to continuously progress regardless of game stage

Comprehensive player interaction

- Allow players to form beneficial agreements
- Implement an enmity system that expands on rivalries between players
- Allow transfers of resources between players

1.0.2 Technology choices

Programming language: C++ 20

Chosen Compiler: MSVC (with Visual Studio 2022)

Rendering Engine: Ogre3D 1.14.2 (not to be confused with Ogre3D-next)

Input Handling Library: SDL2

UI and Overlay Library: Ogre Overlay Component

Concurrency Library: `<threading>` library in C++ included in std libraries

Thread pooling library: `Boost::ThreadPool`

Matrix library: Eigen

JSON parsing library: nlohmann json

Dependency installation and linking: vcpkg library integrated with Visual Studio 2022

Chapter 2: Project Timeline

2.1 Term 1 Timeline

2.1.1 planned timeline

Week 1: write unit tests following TDD and setup build dependencies

Week 2: research efficient ECS and event architecture

Week 3: research input methods for controller and keyboard

Week 4: code central event bus and ECS registry class

Week 5: code input listeners + initial world and instance class

Week 6: code initial split screen functionality and implement event queue

Weeks 7: code entity factories and create entity components for ECS

Weeks 8-9: create initial game map with partial game functionality

Weeks 10-11: prepare interim transcript and improve implementation for presentation

2.1.2 Development timeline

Week 1:

- Research efficient event architecture (moved up from planned Week 2).

Week 2:

- Implement event bus and event queue architecture (significantly ahead of schedule - originally planned for Weeks 4 and 6).

Weeks 3-5:

- Input handling system implementation.
- Achieved reconnection for controllers.
- Automatic device recognition and configuration.

Weeks 5-7:

- Overlay system implementation.
 - Attempted Dear ImGui integration with Ogre3D.
 - Encountered compatibility issues with ApplicationContextBase.
 - Researched and implemented alternative overlay solution (Ogre Overlay).
- Weeks 7-9:** - Config system implementation.

- Universal config system for storing and reading JSON files with nlohmann JSON.

Week 10:

- refinements to implementation.
- thread safety update to event architecture.
- refined input system with more debug information available

Week 11:

- Create ECS registry class with demo implementation (delayed from planned Week 4).
- Create ECS components with ECS factory (delayed from planned Week 7).

Week 12:

- Initial split screen functionality implemented (delayed from planned Week 6).
- Automatic viewport creation on device connection.
- Dynamic viewport resizing.
- Consistent aspect ratio maintenance

Not Completed in Term 1:

- Partial game functionality (originally planned for Weeks 8-9).
- Deferred to Term 2 to allow for REA pattern integration

2.1.3 Retrospective of Term 1s development

Over the development period in Term 1 there were some major differences between the planned timeline and actual development timeline. Specifically,

- Earlier development of event architecture fundamentals
- 2 week delay to implementing an overlay system
- Delay to split screen functionality
- Absence of partial game functionality

Term 1's development has brought to my attention that more careful analysis of planned technologies is required. Specifically, investigating implementations in other projects using similar technologies to me and which prerequisites are required or which existing technology could conflict. Another lesson learned from term 1's development is that more consistent contribution to reports and documentation is required in term 2's development, this way the CFD can easily be used to identify potential race conditions or design flaws.

Despite these delays to the projects timeline, This project has reached a satisfactory end result for term 1. Large architectural groundwork has been completed, making way for development in term 2 along with recognizing REA as a new proven beneficial technology.

2.2 Term 2 Timeline

Week 1: Revisit term 1 implementation, refactor threading model and class communication

Weeks 1-2: Research REA implementations, implement a basic scenario with multiple entities that can perform basic actions.

Weeks 3-4: Create a game world and a basic test scenario where each player is given starting units and starting territory

Weeks 4-5: Model and create simple REA actions for taking territory, attacking units, placing units.

Weeks 6-7: Create an undecorated UI, including context menus for actions that change depending on where the cursor is. settings and main menu screen

Weeks 7-8: Perform market analysis on experienced real time strategy players and implement suggestions as prototypes

Weeks 8-9: Perform user testing with prototypes to produce an unrefined game as the basis for the finished project

Weeks 9-12: Produce a shader complimenting a simplistic art style, implement a refined UI

2.2.1 Development Timeline

Week 1: Refactored threading model using BSthreadPool, implemented per viewport overlays

Week 2: Implemented a TDD scenario for REA, testing attacking scenarios. shell code for the transformation matrix and interaction evaluator

Weeks 3-4: Designed the game globe, added visible cursors to controllers. refactored input translator to allow communication to the render thread

Weeks 4-5: Refactored threading model, implemented per viewport overlays

Weeks 6-7: Created the interaction wheel, created generic functionality for ui elements to be pressed or hovered over.

Weeks 7-8: Performed market analysis via conversations with RTS players, implemented fixes to rendering errors, misaligned button text and a visual representation of selected units

Weeks 8-9: Following Weeks 7-8, I reached out again and implemented unit movement functionality, cities. And blocked users from placing units in the ocean.

Week 10: Implemented REA, added specialised entity components for cities and units. Added a specialised entity loop.

Weeks 10-12: Created a fragment shader responsible for showing enemy units as different colours. Also implemented attacking units functionality. Cities now produce units passively instead of giving the player access to an infinite amount

2.2.2 Retrospective of Term 2s development

Term 2's development followed the planned timeline considerably more closely than Term 1, with the majority of features being implemented within their intended windows. However, a small number of development choices resulted in minor deviations from the original plan. Those being:

- A delay to a working REA implementation
- A delay to producing a playable prototype
- A switch from a visual fidelity shader to a shader responsible for colouring enemy units

The most notable of these was a delay to the REA implementation. Rather than introducing REA alongside the game world in Weeks 3–5 as planned, the decision was made to delay the technology until a functional UI had been created. This was a deliberate trade-off: without any observable visual representation of any entity interactions using REA, validating the correct desired behaviour of entity interactions would have been significantly harder, from both a user testing and development standpoint.

A consequence of this change to the development timeline was that a working prototype wasn't available at the start of the user testing phase. However, valuable feedback was still gathered following early user testing and I was able to respond to this feedback with fixes to the UI and a visual representation of selected units. More feedback was gathered in the following weeks 8-9, bringing about major changes to the final implementation with the addition of cities that passively generate units.

The final major change to the plan for term 2s development was the decision to implement a fragment shader focused on colouring enemy units instead of one focused on visual fidelity. This decision was made due to the development priorities of the project, those being prioritising function over visual fidelity. If I had opted for a fragment shader focused on fidelity then the overall performance of the project would suffer, since colouring enemy units could only otherwise be done with an inefficient CPU heavy loop, thus I opted to solve this problem efficiently with the fragment shader rather than compromise performance.

Despite minor changes to the development timeline, this project has reached a final state consistent with its primary aims. All planned functionality has been completed except one substituted technology, thus leaving this project in a completed end result.

2.2.3 Project aims evaluation

This project has prioritised the primary objectives laid out in interim report, however the secondary objectives haven't been fulfilled in their entirety. The reason for this is a focus on concurrency instead of comprehensive game functionality, thus more time was spent developing the primary focus of this project instead.

This project meets the performance aims, with an fps of 1000 or more on capable hardware, remaining consistent at high entity counts. This project also meets the criteria for local multiplayer functionality, with multiple players being able to play the game at the same time across multiple input devices. A user interface has also been implemented in this project, whilst not fully meeting the specified fidelity standards, the project provides an interaction menu for players to interact with the game.

Chapter 3: Technical Literature Survey

3.1 Core Real time Strategy Game Architecture

3.1.1 Entity Classification and Entity Interactions

RTS games are heavily focused on "units" such as friendly soldiers or a farm that produces food for the player every set amount of time. Each "unit" within an RTS game is highly varied in its behaviour, but may share common attributes such as a position on the screen and a 3D model.

A major challenge faced during the development of RTS games is storing and handling entities differently according to their behaviour; without an efficient method to store and compute the behaviour of large numbers of entities, this causes significant performance concerns as the number of entities increases. Despite there being pre-existing methods to tackle this problem, it is still common to see an "end year" or a "victory year" implemented within RTS games to prevent the number of units from growing too large. The primary reason for RTS games still implementing a limit to game time is the inclusion of highly detailed 3D meshes where they would otherwise be unnecessary.

One popular choice used for entity storage in the development of RTS games is ECS (Entity Component System), since it not only provides efficient storage but also offers the perfect solution for classifying modular entity behaviour through its component system. ECS's component system allows developers to modularise entities into specific behaviours that can easily be reused or removed from an entity to modify its behaviour.

ECS is a powerful choice for real-time strategy games due to entity and component access being constant $O(1)$ time, meaning that the time complexity of managing entities does not increase as more entities are added [5][6]. The reason for ECS achieving $O(1)$ time is its employment of sparse sets — a data structure consisting of two arrays, one sparsely filled with indices and another containing densely packed data. The indices in the sparse array point to entries in the dense array, and this bidirectional relationship is exploited by ECS to achieve $O(1)$ time complexity for both retrieving an entity from a given component and retrieving a component from a given entity. Here is a visual representation of sparse set operation.

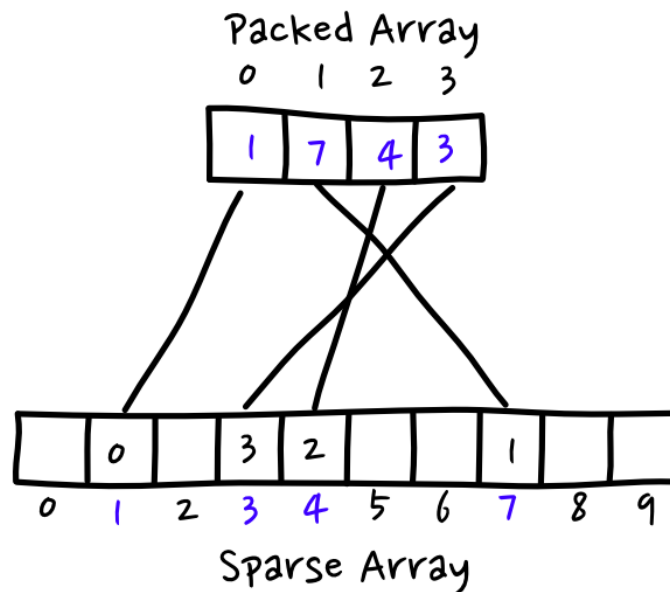


Figure 3.1: sourced from Reference [1]

This diagram, however, does not fully illustrate how ECS implements sparse sets within its component system. Here is a diagram showing what a sparse set in a practical ECS implementation would actually look like.

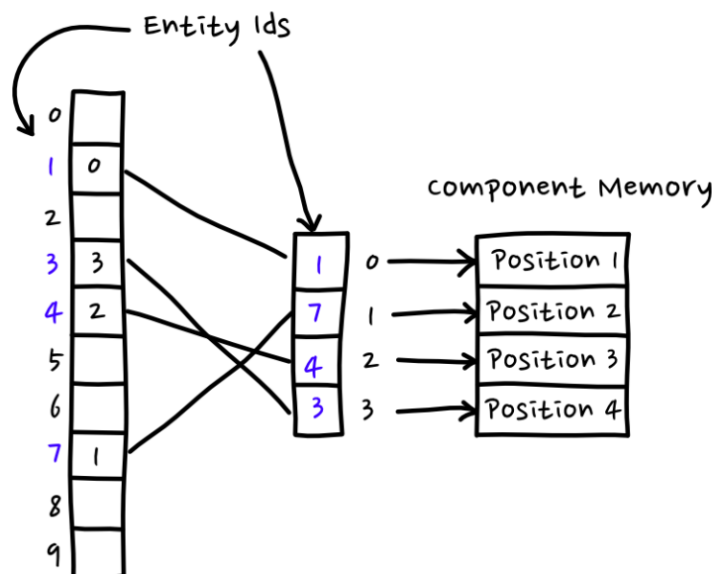


Figure 3.2: sourced from Reference [1]

This diagram shows the relationship between the sparse set and a specific component, demonstrating $O(1)$ lookup time. It is easy to assume that having two components would mean iterating over two different data structures; however, this is not the case — instead, a new sparse set is created for each component [1].

However, evaluating actions between entities remains a significant challenge in RTS games. As the number of units increases, so too does the number of units implicated in any given action, thus exponentially increasing the processing time required.

REA (Resource Entity Action) addresses this issue by modelling entity actions as resource

exchanges, using transformation matrices to evaluate the resulting resources that affected entities hold after a given action. One example where REA proves useful is when two units in an RTS game attack each other — a transformation matrix is used to calculate the resulting health of both entities. The REA pattern has been proven to provide a measurable performance advantage over implementations that do not employ it [2].

As mentioned previously, REA uses a matrix to evaluate actions as resource exchanges, but this is not always appropriate for every entity interaction. REA is primarily suited to routine calculations that are run most frequently and do not involve complex conditions or implications. For instance, an RTS game might use REA to calculate how much currency a factory unit should produce each second, since the conditions governing this are likely straightforward.

3.1.2 Split Screen Multiplayer Approaches

Local cooperative multiplayer (LCM) involves multiple players sharing a single device, with each player receiving an independent portion of the screen controlled by their own input device. LCM is rare in the RTS genre and has seen a general decline across other genres such as FPS and racing, largely due to the expense and complexity of implementation.

A significant technical challenge in LCM is managing multiple input devices simultaneously, as each device must be individually distinguishable and readable in parallel. Games requiring cursor or camera control typically rely on an input library to handle this, the most widely used of which being the GameInput library for Windows and Xbox [4]. SDL2 is a powerful choice for cross-platform compatibility, as it abstracts platform-specific code and keymaps automatically.

Display splitting in LCM is typically handled through fixed-size screen segments, though unified views and distance-based splitting are alternative approaches used by some developers.

3.1.3 How I chose my implementations of ECS

ECS can be implemented via various libraries in C++, two notable ones are Flecs and Entt. Entt uses sparse set to store entities whereas Flecs uses archetype.

Flecs has an advantage over Entt in terms of ease of use due to its query engine, making multiple entity and component retrievals easier to manage. Flecs is also included in the std library in c++ meaning it has the added advantage of not adding extra dependencies to my project.

However the main criteria im concerned with is their efficiency and space overhead. below is a table showing the space and time complexities of both libraries.

Time Complexity

Space Complexity

as can be seen on the table, Entt has the clear advantage in terms of speed and component overhead. whilst Flecs would be significantly easier to use and manage, my final choice for this project was Entt due to the performance focus of this project and any performance edge gained for such a critical architectural feature of this project must take priority.

Operation	EnTT (Sparse Set)	Flecs (Archetype)
<i>Entity Operations</i>		
Create Entity	$O(1)$ amortized	$O(1)$
Delete Entity	$O(1)$ amortized	$O(1)$
<i>Component Operations</i>		
Add Component	$O(1)$ amortized	$O(n)^1$
Remove Component	$O(1)$ amortized	$O(n)^2$
Get Component	$O(1)$	$O(1)$
Has Component (Check)	$O(1)$	$O(1)$
<i>Query/Iteration Operations</i>		
Iterate All Entities	$O(n)$	$O(n)$
View/Query Iteration	$O(n)$	$O(n)^3$
Uncached Query Iteration	N/A	$O(A)^4$
Create Cached Query	N/A	$O(E \log E)$ or $O(E)^5$
Create Uncached Query	N/A	$O(1)$
Lookup Entity by Name	$O(1)$	$O(1)$
Component Registration	$O(1)$	$O(1)$

Table 3.1: Time Complexity Comparison - EnTT vs Flecs sourced using [5] and [6]

Data Structure	EnTT (Sparse Set)	Flecs (Archetype)
Entity Storage	$O(E)^6$	$O(E)^7$
Component Storage	$O(E \cdot C)^8$	$O(E)$ (dense per archetype)
Sparse Index	$O(E)$ per component	N/A (uses archetype structure)
Archetype Storage	N/A	$O(A \cdot C)^9$
Query Cache	N/A	$O(Q \cdot A)^{10}$
Total Memory (avg case)	$O(E + E \cdot C_{avg})$	$O(E + A \cdot C_{avg})$
Total Memory (worst case)	$O(E \cdot C_{max})$	$O(E \cdot C_{max})$

Table 3.2: Space Complexity Comparison - EnTT vs Flecs sourced using [5] and [6]

3.1.4 my use of ECS in this project

ECS entities within this project are created using a factory pattern, where entities being created are split into their individual components then added in serial to the ECS registry to form the complete entity. below is a control flow diagram showing this process for the example cube.

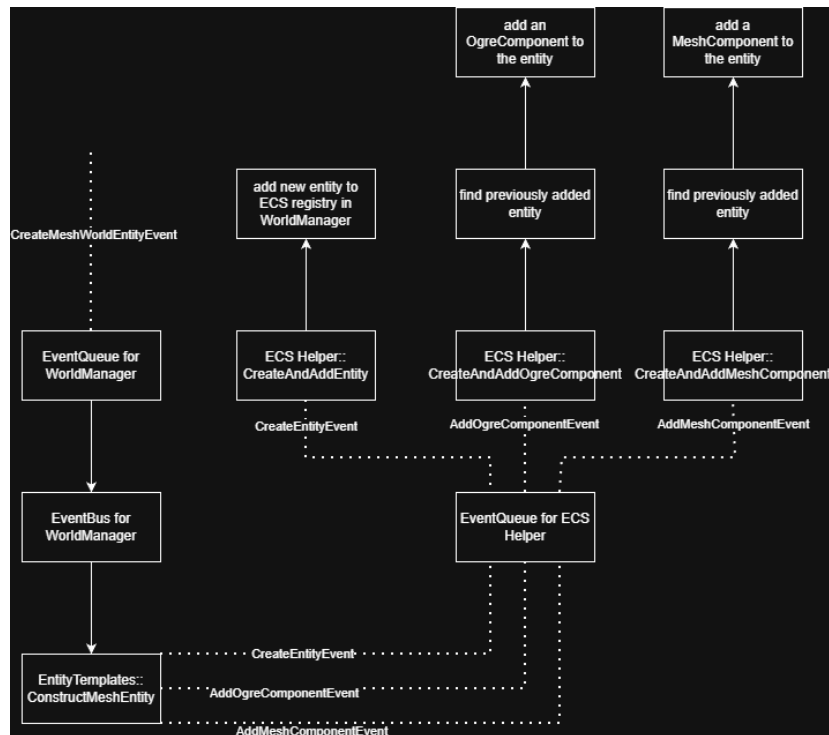


Figure 3.3: a CFD of the example cube creation process

The example cube is made up of 2 ECS components. an `OgreComponent` and a `MeshComponent`, the `OgreComponent` contains an ogre scenenode (a universal object for any ogre entity) and the `MeshComponent` contains the mesh file path.

3.1.5 how i chose my approach to split screen

My chosen approach to split screen in this project is fixed size screen regions. Whilst a unified view of the game world would have been an acceptable choice for this project, i decided against it to prevent confusion with multiple user interfaces appearing simultaneously on the same camera. The lack of individual camera control was also a factor considered, since users wouldn't be able to focus in on specific game events. Fixed screen sizes however do mean that a large display size is required to comfortably play with 4 or more players. This risk can't be mitigated fully, however consistent UI element sizes can assist with readability.

3.2 Concurrency in game Development

3.2.1 threading models

Threading models in game architecture is a fundamental architectural decision when designing games, threading models dictate how processing load is spread across different threads and thus across CPU cores. Since modern CPU's almost always contain more than 1 core with most having 4-8, It's critical that these cores are utilized effectively to improve performance in game design. Single core dependence is a common issue faced in the RTS genre, where only one cpu core is utilized due to architectural limitations, one example of this is *Stellaris* by Paradox Interactive which is known for this issue. Single core dependence is a major concern

for the RTS genre, primarily because processing responsibility for large unit counts is placed solely on one CPU core.

One common approach to threading in game architecture is thread pooling, Thread pools split tasks over a finite number of worker threads through a shared worker queue. One issue with thread pooling however is that two threads may access the same game resource at the same time, requiring careful synchronisation to prevent a race condition. Thread pooling is an extremely common approach in game design, since if it's implemented correctly it can ensure consistent utilization across every CPU core.

Another approach to threading models in games is the actor-entity model where each game object (actors) maintains their own internal state and reacts to messages. This approach avoids race conditions entirely as each thread processes actors without direct memory sharing, meaning two threads will never access the same memory location. This approach does however introduce latency and ordering challenges, messages may arrive late or out of order.

3.2.2 race condition avoidance

Race conditions occur when two different threads attempt to enter the critical region at the same time, e.g two threads access a variable at the same time. Race conditions are problematic since one thread could potentially read the value in the critical region whilst the other thread writes to the variable, causing thread 1's read value of the variable to be incorrect. Race conditions can be avoided in multiple different ways, the most notable being mutex locks where once a thread accesses a critical region it will lock access for other threads until completion. However, this creates problems of its own since one thread will be stuck waiting to enter the critical region and will sit idle until the other thread exits.

Another popular way game developers avoid race conditions is through Event architecture. Event architecture is composed of two main parts, the Event Bus and the Event Queue with the Event Bus notifying subscribing functions immediately after an event is published, then Event Queue's queuing up events to be processed in serial after its dispatch method is called. Event queues avoid race conditions by processing critical region accessing tasks one after the other thus preventing two accesses at the same time.

3.2.3 My approach

My chosen Thread model for this project is thread pooling, each game instance will be assigned a thread that are synced at the end of each frame then reused for the next frame. Due to the nature of Ogre3d only being safely accessible from one thread i have seperated task queues into a local queue and a global queue. the local queue runs thread safe events only accessing that game instances data then the global queue is used to process non thread safe events such as moving a viewport.

Whilst it can be said that Ogre3d only being able to be accessed from a single thread is a limitation of my project, its important to note that down to a fundamental level GPU's run tasks in serial so issuing rendering commands in a multithreaded environment provides no significant advantage.

3.3 Graphics/Rendering Engines

3.3.1 Comparison of commonly used Engines

Game engines are systems that abstract complex rendering from developers. There are a variety of commercial and open source game engines available, each providing differing levels of abstraction. some well known choices being unreal engine and unity which are both considered heavyweight game engines since they abstract processes such as entity storage and concurrency. Open source and lightweight game engines are also widely used, the most popular being GoDot which abstracts considerably less than heavyweight game engines.

Whilst heavyweight game engines are the standard for large game development companies, they do introduce significant overhead and performance implications. The most notable example of this is with Unreal Engines 5's Nanite system for high detail meshes [3], some games use this system unnecessarily introducing no real visual fidelity benefit but causing significant performance degradation. One recent example of this is Borderlands 4, which has been criticized for poor performance on even high end hardware.

Lighter weight game engines are the clear choice when it comes to games focusing on performance. lighter weight game engines run with less overhead and systems such as entity storage are added via stand-alone libraries, following the "pay for what you use" principle. most game developers however still opt for heavyweight game engines despite a lack of real need for them, usually due to ease and thus cost of development.

3.3.2 Overlay and GUI Libraries

Overlay systems are Rendering libraries that render 2d text or shapes over a 3d game world. Overlay systems are used by developers to show debug information such as current input device states and are also used to create GUI's for use by an end user.

One popular Overlay system used in Game development is Dear ImGui. Dear ImGui is widely used to create debug overlays since it abstracts creating text inputs or drop down menus from developers, allowing sophisticated debug overlays to be created easily and conveniently by developers. Dear imgui is however unsuitable for creating a modern GUI for end user's since it's appearance is designed strictly to create tools, in addition to this Dear ImGui redraws the entire ui every frame which is unnecessary for menu buttons that largely remain static.

below is a picture of Dear Imguis UI

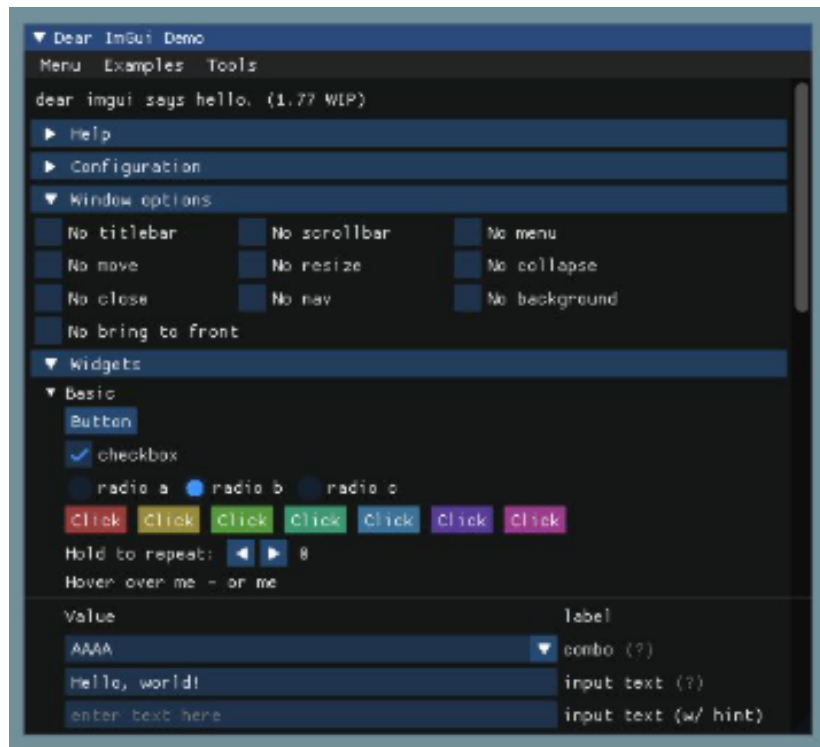


Figure 3.4:

Another popular option is CEGUI, which is primarily used for game GUI's. CEGUI's age and out of date appearance are however criticised, with most game developers opting for a more modern library or creating their own.

3.3.3 My Approach

My chosen Overlay system is the Ogre Overlay library, which is natively included in Ogre3d. Due to Ogre Overlay being part of Ogre3d's ecosystem it allows for convenient integration into this project, this has several benefits such as being able to create unique overlays for each viewport and inclusion within Ogre3d's rendering pipeline. Ogre Overlay is simplistic in terms of features such that Buttons and drop down menus must be created manually, although this means that a GUI library can be produced that is more specialised for this project's needs.

I had initially planned to use Dear ImGui to produce a debug overlay for this project, however due to unforeseen issues with integrating Dear ImGui with ogre i now use Ogre Overlay exclusively.

3.4 Fragment shaders

3.4.1 comparison of commonly used fragment shader technologies

Fragment shaders are a commonly used method for game developers to interact with the graphics pipeline without writing low level ARB assembly programs [14]. Fragment shaders execute for every pixel on screen and are responsible for per-pixel lighting, texture sampling

and visual effects, running millions of times per frame [15]. A key feature of fragment shaders is that they execute solely on the GPU, allowing developers to leverage the massively parallel architecture of modern graphics hardware. This means otherwise CPU-heavy visual tasks can be delegated directly to the GPU, resulting in significantly lower CPU utilisation and a more efficient rendering pipeline overall.

Fragment shaders are written using shading languages, the most common being GLSL, HLSL and MSL. All three are written in a C style syntax, meaning developers familiar with one can switch between them with relative ease, as the differences are largely limited to keywords and platform-specific design philosophies rather than fundamentally different programming models [16]. Despite having similar syntax, the choice of shading language is typically dictated by the target platform rather than developer preference. GLSL is the standard for OpenGL and WebGL applications and is well suited for cross-platform development, HLSL is the primary language for DirectX and Xbox targets and is the default in Unity’s built-in render pipeline, while MSL is exclusive to Apple hardware running Metal [17]. Beyond text-based shading languages, visual tools such as Unity Shader Graph and Unreal’s Material Editor also exist as abstractions to these languages, trading low level control for faster development.

The differences between these approaches become particularly relevant when evaluating which is most appropriate for a given use case, as each carries distinct benefits and drawbacks in terms of platform support, performance control, and flexibility.

Technology	Platform	Syntax Basis	Best For	Key Drawback
GLSL	OpenGL, WebGL, Mobile	C	Cross-platform development	Often compiled differently across platforms causing inconsistent behaviour
HLSL	DirectX, Xbox, Unity	C	Windows and Xbox targets, Unity built-in render pipeline	Lack of cross-platform
MSL (Metal)	iOS, macOS, Apple Silicon	C++14	Performance focused for	Apple exclusive
WGSL	WebGPU	Rust-like	Modern web graphics	Newly introduced with limited functionality
Shader Graph (Unity)	Unity Engine	Node-based (Visual)	Rapid development	Limited low level control

Table 3.3: Comparison of common fragment shader technologies

As can be seen from the above table, GLSL is the only fragment shader language with native cross-platform support, and is purpose built for OpenGL. HLSL, while arguably the more widely used language in commercial game development, is restricted to the DirectX render system and therefore only supports Windows and Xbox, making it unsuitable for a project with cross platform capability. Similarly, MSL is exclusive to Apple’s ecosystem and WGSL uses a non C like syntax. Due to this, the natural choice for a shader language within this project was GLSL, with the added benefit of being natively supported within Ogre3D, my chosen render engine. Ogre3D supports GLSL with its OpenGL render system, allowing fragment programs to be declared and referenced directly within material scripts, meaning parameters such as a player id can be referenced and modified from within the applications code. Furthermore, GLSL’s C style syntax offers ease of development for a project written in C++ due to it removing the need to switch to a more rust like syntax with WGSL.

3.4.2 Common uses for fragment shaders

Fragment shaders play a huge role in modern graphics pipelines, operating at the per pixel level to determine the final colour and appearance of rendered geometry. Fragment shaders where a huge improvement from early fixed-function pipelines which are described as inflexible and lacking the same configurability as fragment shaders [18]. Since then, fragment shaders have allowed developers to explore more complex lighting techniques and post processing effects with fragment shaders being described as the primary way of reaching visual fidelity in real time applications [15][16].

Fragment shaders play a central role in non photorealistic rendering, such as a game using a cartoon art style. A fragment shader could be used to automatically highlight edges of geometry with darkened lines by comparing neighbouring pixels for edge detection. This approach offers a flexible solution to a cartoon art style as fine details such as darkened lines are abstracted from the 3d modelling process, offering ease of development with no compromise to the rendering quality.

One well known and advanced use of a fragment shader is ray marching with a Signed Distance Function (SDF). Where a scene is constructed entirely by mathematical functions instead of a traditional approach where geometry is made up of a list of triangles. An SDF is essentially an advanced implementation of a ray tracing function, returning a signed value of the distance to the closest geometry at a given 3d coordinate, the sign of the value indicates whether the 3d coordinate is inside or outside the geometry [19]. The Ray marching algorithm uses these properties to advance along the ray much like a typical ray tracing algorithm, however the SDF is called at each step to calculate a distance for the ray to skip, without any risk of skipping through any geometry. This process continues until the distance returned by the SDF is under the hit threshold, indicating a hit on a geometries surface.

This technique is used in real time procedural rendering, a technique where geometry is generated instead of loaded from a pre made mesh. In real time procedural rendering, defining geometry as a series of triangle meshes is impractical thus makes a perfect use case for the ray marching algorithm. A key advantage of this approach over traditional rasterisation is that all scene information is inherently accessible during shading via the SDF, whereas in a traditional rasteriser the polygons are processed in isolation with no awareness of other geometry. This limitation of standard rasterisation means that visual effects such as soft shadows must be precomputed and stored in a shadow map [20]. With SDF's, the distance function can be called at any time thus making soft shadows and offer visual effects far more straightforward to compute. This technique can allow the entire rendering pipeline to run through fragment shaders, with no vertex data involved.

Below is an example implementation of the ray marching algorithm in GLSL.

```

1  float rayMarch(vec3 ro, vec3 rd) {
2      float depth = MIN_DIST;
3      for (int i = 0; i < MAX_STEPS; i++) {
4          float dist = sceneSDF(ro + depth * rd);
5          if (dist < EPSILON) return depth;
6          depth += dist;
7          if (depth >= MAX_DIST) break;
8      }
9      return MAX_DIST;
10 }
```

Listing 3.1: A GLSL example of Ray marching in a fragment shader)

The function takes a ray origin `ro` and a ray direction `rd` as inputs. Starting at `MIN_DIST`,

it iteratively advances a depth value along the ray by querying `sceneSDF` at the current position. If the returned distance falls below `EPSILON`, the ray is considered to have hit a surface and the current depth is returned. If the depth reaches `MAX_DIST` without a hit, the maximum distance is returned to indicate a miss. The step size is not fixed, it equals the SDF value at each step. This is what makes sphere tracing efficient: the ray takes large steps through empty space and automatically slows down as it approaches a surface [19].

3.4.3 My approach

GLSL was chosen as the fragment shader language for this project for several reasons. As shown in [table 3.3](#), it is the only shader language with native cross-platform support, which was a core requirement for this project. HLSL was ruled out due to its exclusivity to DirectX and therefore Windows and Xbox only. MSL is similarly restricted to Apple's ecosystem, and WGSL uses a Rust-like syntax rather than C-style, which would feel out of place in a C++ codebase. GLSL's C-style syntax keeps things consistent and reduces context switching during development. It also integrates directly with Ogre3D's OpenGL render system, allowing fragment programs to be declared within material scripts and have parameters like player ID passed in and modified from the application's code.

A simple implementation to display enemy units in different colours was chosen as the fragment shader for this project. Although the ray marching algorithm would have enhanced the visual fidelity, its implementation time would have been unacceptable given the project's focus on concurrency. As a result, a more practical fragment shader was used, prioritising function over visual fidelity. That said, if I were to continue development of this project and eventually publish it then ray marching would be an excellent option to increase this project's visual fidelity.

Chapter 4: implementation

4.1 User interface

A User Interface is a means for players to interact with a program. The user interface for this project reflects the rudimentary focus of the game implementation, with UI elements consisting of basic shapes with overlaid text.

The primary way players interact with the game is through the interaction wheel, a four-button radial menu revealed by pressing R on a keyboard or X on a controller. The interaction wheel was a suitable choice for this project given that the project supports both controllers and traditional keyboard and mouse. Since controllers offer comparatively less precision when moving the in-game cursor, I opted for a design that keeps all interaction options grouped as closely together as possible, reducing the distance the cursor needs to travel regardless of input method.

4.1.1 User interface visibility across viewports

The user interface was built using OGRE3D's overlay system, a 2D rendering layer included within OGRE's ecosystem that sits on top of the 3D viewport. Since overlays are tightly coupled to OGRE3D's rendering pipeline, all creation and management of overlay elements is handled exclusively on the render thread. Despite this constraint, a clear separation between UI elements is maintained by mapping them into distinct `Ogre::Overlay` instances based on the viewport they belong to. This is done using an `std::unordered map` with the `Ogre::Overlay` as the key and the `Ogre::Viewport` as the value, allowing for $O(1)$ retrievals of Owned viewports.

This separation of overlay elements allows for an override of OGRE3D's `PreViewportUpdate`, a listener method called just before a viewport is rendered. Within this override, the overlay corresponding to the active viewport is made visible while all others are hidden, ensuring each overlay is only ever rendered to its intended viewport. This approach avoids the need for an extensive ownership check on the `Ogre::Overlay` and keeps the UI logic self-contained within the render thread.

```

1  // disable all the overlays, then enable the one that should be rendered
2  for (auto Viewport : InstanceViewports) {
3      if (Viewport.second->Equals(evt.source)) {
4          Viewport.first->show();
5          char IdChar = Viewport.first->getName().back();
6          PlayerId = IdChar - '0';
7      } else {
8          Viewport.first->hide();
9      }
10 }
```

Listing 4.1: iterating over stored the unordered map of viewports and overlays to show or hide accordingly (`Product.sln`, `ViewPortUpdateListener.cpp`)

The `PreViewportUpdate` override receives a `RenderTargetViewportEvent` containing a reference to the active viewport, which is used to identify which overlay should be shown. The correct viewport is then identified by iterating over the `std::unordered map` where each

one is checked against the active viewport from `RenderTargetViewportEvent`. Once the correct overlay is determined, `setVisible(true)` is called on it while the remaining overlays are hidden by calling `setVisible(false)`, ensuring no overlay is shown on an unintended viewport.

4.1.2 User interface interaction

With the overlay visibility system in place, the next challenge was implementing interactive UI elements within those overlays. OGRE3D's overlay system provides the basis for this through `Ogre::OverlayElement` and its subclasses, which represent individual UI components such as panels and text areas. However, OGRE3D's overlay system provides no built-in functionality for interactivity such as click detection or hover states, meaning any interactive behaviour had to be implemented manually.

The first challenge with implementing interactive UI elements was that `OverlayController` runs on the render thread, whilst cursor positions from input devices are evaluated inside the instance threads. The solution to this was to propagate cursor positions from the instance threads to `OverlayController` via a `MouseMoveEvent`, which carries both the identifier of the input device that moved and the viewport it belongs to. Using this information, `OverlayController` is able to narrow the cursor check down to the specific `Ogre::Overlay` associated with that viewport, rather than checking every overlay on every cursor movement. The relevant overlay is then retrieved by name, after which each contained `Ogre::OverlayElement` is iterated over and `findElementAt` is called to determine whether the cursor is positioned over it.

```

1 void OverlayController::OverlayCursorCheck(CursorMovementEvent Event) {
2     RenderSystem& RS = RenderSystem::GetInstance();
3     ViewPortController* EventVPC = RS.FindViewPortFromDevice(Event.Device);
4
5     // find the overlay specific to the device being moved (as to not check
6     // everything for each device)
7     std::string OverlayName = "UI_Overlay_" + std::to_string(Event.ThreadNumber
8     );
9
10    Ogre::Overlay* SpecificOverlay = OverlayMgr->getByName(OverlayName);
11
12
13    Ogre::Overlay::OverlayContainerList ContainedElements =
14    SpecificOverlay->get2DElements();
15
16    for (Ogre::OverlayElement* Element : ContainedElements) {
17        OverlayInfo* Info = nullptr;
18        try {
19            Info = OverlayInfos.at(Element->getName());
20        } catch (std::exception e) {
21            OverlayErrorReporter.EnqueueError(ErrorDetail::CreateError(ErrorCode::
22            OVERLAY_MISSING_INFO));
23            return;
24        }
25
26        if (Element->findElementAt(Event.RelativeXY[0], Event.RelativeXY[1])) {
27            Info->Hovered = true;
28            //pressing the overlay should always override hovering
29            if (!Info->Pressed) {
30                OverlayHovered(Element, Info);
31            }
32        }
33    }
34}

```



```
28     } else {  
29         Info->Hovered = false;  
30         OverlayReleased(Element, Info);  
31     }  
32 }  
33 }
```

Listing 4.2: performing a cursor check on elements contained within an Overlay (Product.sln, OverlayController.cpp)

The next challenge was implementing UI elements that had an effect when pressed. My initial approach to this problem was to store a pointer to the buttons state, either true or false indicating whether the button was pressed. A large issue with this approach was the lack of thread safety, `OverlayController` may write true to the value on one tick then false on another causing the instance thread to completely miss the trigger window where the value is true. The final chosen approach to this problem was a callback system, once the button had been pressed `OverlayController` calls a stored lambda callback to trigger the desired effect. For example, the main menu "play" button is registered with a lambda to queue a state change event, thus once the same tick that the button is pressed the application changes state with no chance of the instance thread not recognising it.

4.2 concurrency

This project has implemented concurrent execution of each game instance, with each instance managed using the thread pooling pattern. `BSThreadPool` is the chosen library for achieving this functionality, selected for its straightforward integration and ease of development. In this project, each player is assigned their own independent game instance, and `BSThreadPool` manages the run method for each of these game instance threads automatically. Rather than constantly creating new threads, threads are recycled each frame, with new threads only being created when a new player connects.

Some concurrent aspects of the project do not however run within the thread pooling pattern, instead occupying dedicated standalone threads to reflect their independent, continuous nature. The input listener is one such example, running persistently on its own thread to ensure responsiveness is never delayed by game instance workload. Across the standalone threads and the thread pool workers, the implementation is designed to occupy a total of 4 threads, with the intention of leaving 4 spare cores on an 8-core CPU for game instances. Some of these independent threads such as Input analyser take up a very small amount of cpu resources, with 12 percent usage being recorded, allowing background OS tasks to not be disrupted by the projects operation.

4.2.1 My race condition avoidance approach

A race condition occurs when two threads running concurrently attempt to access the same shared resource at the same time, with at least one thread writing to that resource. Race conditions produce unexpected behaviour in multi-threaded programs, so prevention was a priority throughout the development of this project.

This project negates race conditions using event architecture, supported by the addition of event queues. All shared resource access is controlled via events, which act either as a request to complete a task or as an asynchronous information request. A typical example of this can

be seen in Listing 4.4, which demonstrates how events are used to prevent a race condition when issuing a ray trace command.

```

1  void PlayerGeneralControl::OnPress(PressActionCommand Cmd) {
2      RenderSystem& RS = RenderSystem::GetInstance();
3      std::vector<float> Position =
4      std::vector<float>{Cmd.Context.MouseX, Cmd.Context.MouseY};
5      RS.RenderQueue->Enqueue(StartRayTraceEvent(
6      Position, PlayerTranslator->ManagedDevice,
7      [](EventQueue* queue, EndRayTraceResultEvent Event) {
8          queue->Enqueue(Event);
9      },
10     TriggerQueue));
11 }

```

Listing 4.3: Issuing a `StartRayTraceEvent` to the Render Queue to avoid direct cross-thread access to Ogre3D (`Product.sln`, `PlayerGeneralControl.cpp`)

This code runs inside an instance thread and handles issuing a ray trace command to Ogre3D. If this code were to access Ogre3D directly, a race condition could occur if a separate thread accessed Ogre3D at the same time. Instead, a `StartRayTraceEvent` is issued to the Render Queue (the centralised queue for all Ogre3D interactions) and processed in serial with other Ogre3D interactions, preventing any direct cross-thread access.

A further element of race condition avoidance introduced in the second term of development is the use of a lambda callback for information requests. Since events are not processed immediately after being queued, it became necessary to find a solution to the absence of immediate information retrieval. As shown in Listing 4.4, a lambda is created to queue an `EndRayTraceResultEvent` back to the sender, ensuring that `PlayerGeneralControl` is notified of the ray trace result in a thread safe manner.

This Approach to Race condition avoidance allows this project to remain lock free for the most part, making this project easily extendable. The only method directly accessed across threads is the `Enqueue` method in the event queue class, for this reason its necessary to use a mutex as seen below.

```

1  template <typename EventType>
2  void Enqueue(EventType&& Event) {
3      std::lock_guard<std::mutex> Lock(QueueMutex);
4      Queue.push([Event = std::forward<EventType>(Event)](EventBus& Bus) mutable
5          {
6              Bus.Publish(std::move(Event));
7          });
8  }

```

Listing 4.4: Issuing a `StartRayTraceEvent` to the Render Queue to avoid direct cross-thread access to Ogre3D (`Product.sln`, `PlayerGeneralControl.cpp`)

The `Enqueue` method can take any Event as a parameter as signified by the template keyword, this allows any event to be added to the queue, without the need to create hundreds of `Enqueue` methods for each event type.

4.3 Resource Entity Action

A Major technology implemented throughout term 2's development was Resource Entity Action, a methodology using a matrix to evaluate entity interactions much faster than standard arithmetic.

Since this project is focused mainly on a rudimentary implementation of a game the only practical implementation of REA used is between two entities attacking eachother. As seen in the below snippet, a matrix is used to calculate the resulting HP (health points) of the defending entity.

```

1 Eigen::MatrixXf CombatMatrix(2, 2);
2 CombatMatrix << 0.0f, 0.0f, -1.0f, 0.0f;
3 void EntityInteractionEvaluator::ProcessAttackEvent(AttackEvent Event) {
4     Eigen::VectorXf Resource(2);
5     Resource << (Event.AttackPower), (*Event.DefendingEntityHP);
6
7     AttackMatrix->EvaluateAction(Resource, Event.DeltaTime);
8
9     *Event.DefendingEntityHP = Resource(1);
10 }

```

Listing 4.5: evaluation of an attack event using an eigen matrix (`Product.sln`, `EntityInteractionEvaluator.cpp`)

The combat matrix used to evaluate the resulting defender HP is defined using Eigen as a 2x2 vector. The matrix definition is designed to evaluate the defenders hp with $d(HP)/dt$ which equals $-1 \cdot AP + 0 \cdot HP = -AP$, causing a decrease in defender hp proportional to the attack power, additionally scaled by delta time for consistency. The use of a matrix such as this one to evaluate attack events is significantly faster than standard arithmetic, whilst this project doesn't utilise matrix's enough to make a comment about its speed, its been found to increase performance significantly [21].

4.4 input handling

This project has implemented Input handling using SDL2 with concurrent input translation happening for each game instance. both controller and keyboard+mouse input is read, both types of input device's activity is translated to a cursor position on the screen and a camera angle (changes based on relative movement).

Another implemented feature for input handling in this project is disconnect handling. if a controller is disconnected whilst the project is running, then the controller is reconnected to the correct game instance once the device reconnects. In addition to this, a new input device can be connected, causing another game instance to be created with its own viewport.

However, the disconnect handling is limited by a known issue with SDL2, generic controllers cant be distinguished on the basis of vendor and product id. This limitation prevents truly accurate reconnect handling as soon as 2 or more controllers disconnect, most games solve this using a text prompt, but I decided against this solution as a faulty controller could cause a repeated prompt.

4.5 ECS

This project has implemented ECS for storage and classification of game objects. ECS entities can easily be created using entity templates, these entity templates define common game objects such as units and cities, adding relevant components such as `ProducesUnitsComponent` or `AttackComponent` to define their behaviour.

ECS entities in the final version of this project also have common entity components, such as `Ownership` component, allowing common information to be readily available on demand. These common components don't define any behaviour, but eliminate the need for any context queries surrounding a given entity, increasing the efficiency of iterating over these entities.

4.5.1 Use of ECS within the final version

The entity loop

Every implementation of a game requires an entity loop to evaluate entity behaviour each tick. Most games achieve this using a singular $O(n)$ loop that evaluates every entity per tick regardless of context. This approach introduces concerns around both readability and efficiency, as each entity is evaluated against every possible behaviour even when only a small subset applies to it. These unnecessary checks increase performance overhead and reduce code readability, making future development and maintenance considerably more difficult. In more advanced scenarios where entity counts grows significantly, this overhead increases considerably and is difficult to solve without a full restructure of the entity loop [2].

The use of ECS within this project allows for an alternative and cleaner approach to the entity loop [1]. Because entities are categorised by their stored components, the entity loop can be structured modularly, where each system only queries and processes the entities relevant to it. The ECS library chosen for this project was `EnTT`, which uses a sparse set data structure internally. This allows registry views to efficiently retrieve all entities qualifying for a given behaviour check, so that, for example, only entities possessing an `AttackComponent` are evaluated when processing attack logic. This targeted querying eliminates the redundant checks present in a traditional $O(n)$ loop and produces a far more readable system structure where each behaviour is handled in isolation. Notably, `EnTT`'s sparse set architecture has been implemented in commercial game titles, with `Minecraft: Bedrock Edition` being a well documented example [5].

The choice of a sparse set ECS over an archetype based ECS carries meaningful performance implications worth considering. Research comparing the two approaches finds that sparse set implementations offer cheaper entity modification costs, making them well suited to scenarios where component composition changes frequently at runtime [5]. Archetype based systems, by contrast, store entities of the same component composition contiguously in memory, which improves cache coherence during large scale iteration but incurs a higher cost when an entity's composition changes [5][6]. For this project, where entities regularly gain and lose components as the game evolves, the sparse set model offered the more practical trade-off. While an archetype approach may offer iteration advantages at very large entity counts, the flexibility of `EnTT`'s sparse set model was better aligned with the dynamic nature of this project's entity behaviour.

```

1 for (auto& [PlayerID, Entities] : PlayerProductionMap) {
2     for (entt::entity Entity : Entities) {
3         auto& Produces = View.get<ProducesUnitsComponent>(Entity);
4         auto& Ownership = View.get<OwnershipComponent>(Entity);

```

```

5
6     const float ProgressPerSecond = (Produces.NumPerMinute / 60.f) * 100.f;
7     Ownership.GamePlayer->PlayerQueue->Enqueue(
8     UpdateUnitProgressEvent(ProgressPerSecond * DT));
9 }
10 }

```

Listing 4.6: a small section of the entity loop (Product.sln, WorldManager.cpp)

4.6 The running states of the project

4.6.1 Term 1

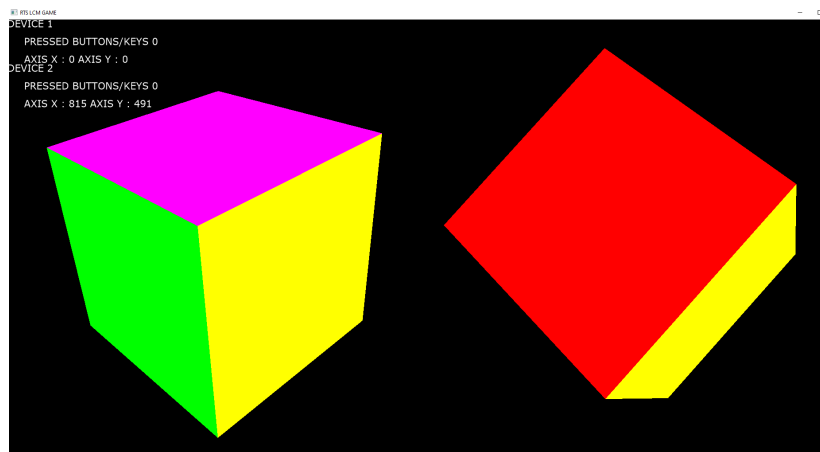


Figure 4.1: a screenshot taken of the projects execution

This is the visible state of the project when 2 input devices are connected. both input devices have their own perspective of the cube and can control their perspective simultaneously. the debug overlay also shows the current state of each input device such as the number of pressed buttons.

4.6.2 Term 2

This is the running state of the project in Term 2. As can be seen the world map is represented as a globe, with raised land masses representing countries.

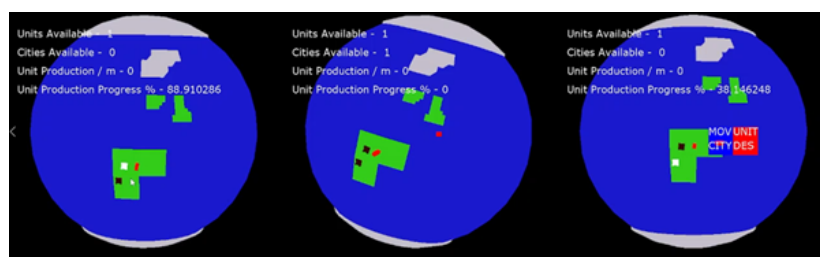


Figure 4.2: a screenshot taken of the running state of the project in term 2

A 4 button interaction menu can be seen in the above screenshot, which is the primary way users interact with the project. From the interaction menu users can either move a selected unit, destroy a selected unit, place a city or place a unit. Since this project focus is mainly on concurrency rather than visual fidelity, the UI elements remain basic along with the main games UI. That being said, the UI elements are fully interactive, giving an indication of whether they're hovered or pressed.

4.7 Locking interaction and visibility between viewports

4.7.1 Preventing controlling other viewports UI elements

Given the local cooperative nature of this project, it was necessary to implement safeguards preventing players from inadvertently or deliberately taking control of another player's viewport.

For game controllers, this is a relatively straightforward problem to solve. Since controllers do not produce an OS-level cursor, an artificial cursor can be created programmatically and its coordinates clamped to the boundaries of the assigned viewport, effectively restricting each controller to its designated region of the screen.

The mouse cursor presents a more complex challenge. Unlike a controller's artificial cursor, the OS cursor cannot be locked to a portion of the screen without introducing significant inconvenience to the user, for instance, a player may need to interact with system-level elements outside the application. As a result, free cursor movement was permitted for this project.

```

1   Ogre::Overlay* SpecificOverlay = OverlayMgr->getByName(OverlayName);
2
3   Ogre::Overlay::OverlayContainerList ContainedElements =
4   SpecificOverlay->get2DElements();
5
6   for (Ogre::OverlayElement* Element : ContainedElements) {
7       OverlayInfo* Info = nullptr;
8       try {
9           Info = OverlayInfos.at(Element->getName());
10      } catch (std::exception e) {
11          OverlayErrorReporter.EnqueueError(
12              ErrorDetail::CreateError(ErrorCode::OVERLAY_MISSING_INFO));
13          return;
14      }
15
16
17      if (Element->findElementAt(Cntxt.MouseX, Cntxt.MouseY) && !Cmd.Released) {
18          Info->Pressed = true;
19          OverlayPressed(Element, Info);
20      } else {
21          Info->Pressed = false;
22          if (Info->Hovered) {
23              OverlayHovered(Element, Info);
24          }
25      }
26  }
```

Listing 4.7: accessing a viewports owned overlay elements and excluding unowned when checking if they were pressed (`Product.sln`, `OverlayController.cpp`)

As shown above, when a mouse click or trigger input is received, only the overlay elements belonging to the active viewport's assigned overlay are iterated over. This is made possible by each viewport maintaining its own dedicated overlay within Ogre3D, which can be retrieved by name to produce a list of exclusively owned overlay elements. This design offers two key benefits: it eliminates redundant checks across unrelated overlay elements, and it inherently prevents any input device from interacting with elements outside of its assigned viewport, naturally enforcing the input boundaries discussed previously.

4.7.2 Showing enemy units as a different colour

Within the real time strategy genre, it's necessary for other players to be distinguishable from one another. Most RTS games achieve this through a unique colour for each player with borders and a name tag to indicate a country. However, due to the simplicity of the game implementation of this project there's no need for a unique colour assigned to each player and a typical player would only be concerned with having their own units be distinguishable from other players units. Therefore the chosen colour scheme for this project is white for the player themselves and red for other players.

Showing other units as different colours introduces significant technical challenge. My initial thinking to solving this problem was to iterate over each entity before they're displayed on a viewport, then change their colour dynamically to reflect friendly or enemy units. This solution is however extremely inefficient since it adds an $O(n)$ loop for each game instance, causing the render thread to slow down significantly.

My chosen approach to this problem was glsl shader program as seen below.

```

1  #version 330 core
2  uniform float viewingPlayerID;
3  uniform float ownerID;
4  void main()
5  {
6      if (abs(ownerID - viewingPlayerID) < 0.5)
7      {
8          gl_FragColor = vec4(1.0, 1.0, 1.0, 1.0);
9      }
10     else
11     {
12         gl_FragColor = vec4(0.2, 0.04, 0.04, 1.0);
13     }
14 }
```

Listing 4.8: glsl shader that changes a mesh's material dynamically based on the mesh's owned player id and the viewing player id (`Resources/world/mat, unit fp.glsl`)

Rather than iterating over entities on the CPU each tick to update material states, this implementation delegates the colour logic directly to the graphics pipeline using a GLSL fragment shader. GLSL (OpenGL Shading Language) is a C-like high-level shading language that runs natively on the GPU, allowing logic to execute in parallel across thousands of cores

without involving the render thread [14]. In Ogre3D, material scripts can reference GLSL programs directly, meaning this approach is not only supported natively by this project's dependencies, but also complements its concurrency-focused architecture.

This approach is considerably more suitable than iterating over entities in `preViewportUpdate`. The loop-based alternative requires `setMaterial()` to be called per entity, per tick, on the render thread — each call capable of triggering shader program rebinding, texture unit rebinding, and GPU state flushing [15]. This places the full cost of the ownership condition on the CPU, stalling the generation of the next frame until each entity is evaluated. The GLSL approach removes this responsibility from the render thread entirely, with the ownership condition resolved solely by the GPU.

A further benefit of this approach is scalability. The loop-based approach degrades linearly as entity count increases, since each additional entity causes another `setMaterial()` call per tick. With the fragment shader, the GPU resolves the `ownerID` and `viewingPlayerID` comparison across all relevant fragments simultaneously during rasterisation [14], meaning the CPU cost remains effectively constant regardless of how many entities are present.

Chapter 5: performance comparisons

5.1 multi-threaded vs single-threaded

Since the primary focus of the project is to produce a multi-threaded game I have decided to outline the relative benefit that has been achieved through using concurrency. Below is a screenshot of the frames per second recorded in the program for the multi threaded implementation.

A screenshot of a black rectangular box with white text that reads "FPS - 2680.9653".

Figure 5.1: a screenshot taken of the frames per second counter in the standard multi threaded mode

Below is a screenshot from a converted implementation where everything runs in the main thread.

A screenshot of a black rectangular box with white text that reads "FPS - 1194.6005".

Figure 5.2: a screenshot taken of the frames per second counter in the converted single threaded mode

There are, however, some important limitations of this comparison to consider. The most fundamental is that this project was designed specifically around multi-threading, meaning the event architecture approach is entirely inappropriate for a single threaded environment and inherently favours multi-threaded execution. Event architecture with event queues introduces additional overhead in a single threaded conversion, since events are still enqueued and dispatched despite these steps being entirely unnecessary in a single threaded context where no race conditions can occur. This overhead penalises the single threaded version in a way that is structurally unfair, as it measures the cost of a concurrency solution in an environment where concurrency is not present.

A further limitation is that a large number of objects within this project are allocated to the heap. This is a deliberate design decision in the multi-threaded implementation, as heap allocation allows for cleaner ownership and lifetime management across threads. In a single threaded environment, however, this becomes unnecessary. Classes that currently operate on instance threads, such as `InputListener`, could instead be processed within the same call stack, allowing stack allocation or direct in-place processing, both of which carry considerably less overhead than repeated heap allocation.

A third limitation concerns thread overhead itself. NVIDIA's technical documentation notes that many CPU-bound applications degrade in performance as thread count increases, citing hardware resource contention, increased pressure on the memory subsystem, and higher rates of OS context switching as contributing factors [7]. The single threaded version of this project introduces exactly these kinds of artefacts via the event architecture without receiving any of

the parallelism benefits that justify them, further skewing the comparison against the single threaded implementation.

Taken together, these limitations mean that a fair and meaningful comparison between a multi-threaded and single threaded version of this project would require a ground-up redesign of the single threaded implementation to better suit that execution model. This would involve replacing the event queue architecture with direct function calls, reducing unnecessary heap allocation, and restructuring subsystems such as `InputListener` to operate without thread boundaries. Such a redesign would be too time consuming given the time constraints of this project and falls outside its intended scope.

Chapter 6: deployment

6.1 deployment prerequisites

at a bare minimum you should possess:

- an x64 based cpu - an operating system compatible with OpenGL or DirectX - at least one games controller to test the split screen functionality - a modern dedicated GPU, preferably one with more than 4gb of video memory.

additionally, before deploying this project please ensure you have installed and configured the following:

- Visual studio Community edition 2022 - the vcpkg component of visual studio

details on how to install and configure the above can be found in the deployment video provided in the git repository.

6.2 deployment instructions

- * Open product.sln (if a retarget prompt appears then refer to deployment prerequisites)
- * This project uses cplint in pre build events, run "pip install cplint" (important that this is done in the vs python command prompt and not a regular command prompt)
- * You'll be presented with 3 different projects in visual studio, "product" is a static lib and "tests" + "application" is an exe
- * Before compiling, configure "multiple startup projects" and make sure that the static lib isn't run but the other 2 are
- * Clean then rebuild all using visual studio, vcpkg will download and install dependencies (usually takes 2- 3 hours)
- * Please note that if the first run fails it means that vcpkg automatic linking failed. as a fallback, post build events will copy all required DLLs to the output dir meaning the second run wont fail.
- * Please note that this project requires OpenGL to be installed, the glsl shader will not function when this project is used with DirectX * ***VERY IMPORTANT*** if you are running my project from inside visual studio you must change the debugging directory (working directory in the debugging section) for each individual project (application, product, tests) to \$(OutDir) (must be uniform between projects), prebuild events will copy the right ogre config to the outdir so please keep in mind if you decide to build my project to a different directory (outside of visual studio) then you need to recreate this behaviour.

* ***VERY IMPORTANT*** you must ensure that the pre and post build events have been preserved after cloning this repository. Without these, the dependancy DLLs and resource folders wont be copied to the output directory.

6.3 important note for linux

This project was designed fully around platform-neutral code. That being said, due to my use of visual studio as my preferred IDE the deployment process is very much intended for windows. This is due to the added time consumption of maintaining a CMAKE compiled version alongside the primary visual studio version, thus the intended way to run this project on linux is through proton (valve compatibility layer) on a pre compiled executable.

Chapter 7: Software engineering methodologies

7.1 testing environment

This project uses the Doctest C++ library for unit testing with Cpplint to check for google c++ style compliance. To test for memory leaks and concurrent performance within my project im using the native visual studio debugger with Open Hardware Monitor to check core utilization.

This project is primarily tested on a computer running windows 11 and one running linux mint (through proton).

Primary development computer

Operating System: Windows 11 Home edition

CPU: Ryzen 9 9950X3D

GPU: Nvidia RTX 5090

RAM: 64GB DDR4

disk running and storing project: Nvme SSD

secondary development computer

Operating System: Linux Mint with proton installed

CPU: Ryzen 7 3700x

GPU: AMD Radeon 7700XT

RAM: 16GB DDR4

disk running and storing project: standard sata SSD

both running at factory clock rates

7.2 Use of TDD

This project was developed using a structured and methodical approach to modern software engineering practices. using industry standard system design from both a Game development and a concurrency standpoint, My Deployment process is also streamlined for rapid deployment with unit testing done to ensure correctness. Unit tests are automatically run alongside this project when run inside visual studio meaning i can easily identify if a new feature has affected previous functionality, alongside this cpplint is run as a prebuild event to check for google c++ standard compliance.

7.3 system architecture

This project uses Event architecture and Entity Component System, an industry standard pattern for game development. Event architecture has allowed this project to reduce class coupling since classes communicate via events instead of direct access. Event queues also

manage race conditions since events are processed in serial rather than having two conflicting events running in parallel.

Another architectural standard this project follows is Thread pooling, this is an industry standard approach to concurrency as it allows consistent hardware utilization across cpu cores.

7.4 Version control

This project uses Git as a version control solution, with the Git repository being hosted on the department git lab. Progress was frequently committed throughout the development process, and the main branch was kept in a working state throughout to keep a restore point available. Major changes to this project were created on separate branches with branch objectives set before hand, before merging into main the feature branch is thoroughly tested and checked against branch objectives.

Chapter 8: **Acronomys**

ECS - Entity Component System
REA - Resource Entity Action
RTS - Real Time Strategy
SDL2 - Software Direct Media Layer
LCM - Local Multi Player
CPU - Central Processing Unit
GPU - Graphical Processing Unit
RAM - Random Access Memory
TDD - Test Driven Development
GUI - Graphical user interface
GUI - Graphical user interface
GLSL - OpenGL shading language
HLSL - High level shading language
MSL - Metal shading language
WGSL - WebGPU shading language

Chapter 9: Glossary

Ogre3d - a lightweight rendering engine in the form of a c++ library

Ogre Overlay - a component included inside the Ogre3d ecosystem, focused on overlaying 2d objects over a 3d viewport

Dear Imgui - a c++ rendering library focused on creating debug overlays

Software Direct Media Layer - a commonly used cross platform input handling library for c++

Resource Entity Action - a design pattern for real time strategy games that models entity interactions into transformation matrices that define resource exchanges between entities for fast evaluation

Entity Component System - a software architecture pattern that separates game objects into entities (identifiers) and components (pure data containers). ECS is used where composition is prioritized over inheritance

entt - an Entity Component System library for C++ that uses sparse set to store entities

nlohmann json - a popular JSON library in c++

stellaris a real time strategy game from Paradox interactive, known to have performance issues

Fragment shader A programmable stage of the rendering pipelines, running for each pixel on the screen

Ray tracing A methodology for computing the distance from a 3d origin point to other geometry, in a specified direction

Eigen A c++ library for defining and performing matrix calculations

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Chapter 11: appendices

11.1 project diary

11/10/25 - researched convenient way to implement TDD
 11/10/25 - configured and successfully setup project
 12/10/25 - configured cpplint to check c++ google standard
 12/10/25 - designed general structure of ECS Registry and Event bus
 12/10/25 - created initial test templates
 12/10/25 - created EventBus and EventQueue tests
 13/10/25 - researched ogre render system patterns
 13/10/25 - initial ViewPortController and RenderSystem classes and tests
 14/10/25 - investigated the best way of harvesting input simultaneously from multiple controllers (arrived at SDL2, integrated into ogre by default)
 15/10/25 - created initial event templates and InputListener class
 15/10/25 - shifted focus to ECS storage, researched how entt registry should be managed and who should own it. decided on a WorldManager singleton
 16/10/25 to 19/10/25 - modeling runtime scripts and researching other ogre3d projects
 20/10/25 - race condition tests, initial implementation of runtime scripts using app states
 21/10/25 - write base renderSystem class, designed error and logging system
 22/10/25 - wrote EventBus and EventQueue class as prerequisite to error and logging system
 22/10/25 - realised EventQueue class erases type information, redesigned and placed queue items in a wrapper class
 24/10/25 - wrote passing tests for eventQueue
 25/10/25 - when doing ogre runtime tests i realised that the vcpkg version of ogre is incorrect, porting fully to a manual build
 26/10/25 - identified that the issue was due to string corruption of the package name, ported back to vcpkg and fixed by linking debug libraries when necessary
 27/10/25 - wrote passing viewport tests (apart from an unidentified issue regarding viewport z-orders)
 28/10/25 - passing tests across the board (bar 2 viewport tests), created a responsive ogre render window
 29/10/25 - integrated and initialised SDL2 and Dear Imgui
 30/10/25 - partial implementation of error reporting (without handling yet)
 30/10/25 - looked at sample programs in the ogre and sdl doc, realised input listener cant be on a per instance basis, redesigned such that a singular input handler dispatches to each instance
 31 4/11/25 - which data structure and key is appropriate for having an input device bound to an sdl id? (same issue for input device and queue)
 4/11/25 - read <https://www.geeksforgeeks.org/cpp/map-vs-unordered-map-c/>
 4/11/25 - read <https://www.geeksforgeeks.org/cpp/set-vs-unordered-set-c-stl/>
 4/11/25 - read <https://www.geeksforgeeks.org/cpp/advantages-of-vector-over-array-in-c/>
 4/11/25 - settled on unordered set instead of vectors, mainly for O(1) lookups but also since its a set and it only needs to store unique data
 5/11/25 - how can i replicate case sensitivity for sdl when sdl2 doesnt generate capital letter events?
 5/11/25 - read <https://discourse.libsdl.org/t/case-sensitive-key-down/10807>
 5/11/25 - realised sdl2 has global "mod" events, used them to check for shift and capslock state to replicate case sensitivity
 6/11/25 - test port to secondary pc
 6/11/25 - realised that plugins.cfg used by ogre reads exclusively from the build output dir

which isnt replicated on gitlab

6/11/25 - added a prebuild event to copy from product/ to (Outdir)

7/11/25 - how can i add specific error information for my error handling?

7/11/25 - used previously familiar format library

7/11/25 - encountered issue where format would read string values at compile time as const

7/11/25 - realized limitation of format library with a static lib and researched alternatives

7/11/25 - read <https://stackoverflow.com/questions/76329898/fmt-compile-time-format-string-check-without-generating-asm-code-for-the-check>

7/11/25 - implemented fmt library instead as dependancy then used fmt in CreateError calls to force runtime string building

8/11/25 - researched ImGui implementation and consulted ogre doc with imgui example at <https://github.com/OGRECave/ogre/tree/master/Samples/Simple/include>

8/11/25 - realized this example didnt apply to me due to their use of ApplicationBase (used to make cookie cutter ogre3d apps not relevant to me)

8/11/25 - attempted custom implementation of imgui in my project, repeated failure

8/11/25 - consulted countless forum posts on the ogre forum, many outdated and irrelevant to the issue

8/11/25 - realized imguis IO string fields "BackendRenderer" was corrupt/inaccessible

9/11/25 - realized that im building imgui and ogre as dynamic library with no way of accessing a shared IO

9/11/25 - edited project properties to build as static lib (for unified memory)

9/11/25 - major edits to project setup for compatibility with static lib versions of ogre and sdl2 etc

10/11/25 - using ogre in static lib mode proved to be a major pain, reverting back to last stable build

10/11/25 - after revert, used native ogre overlay support to add OverlayController (class owned by RenderSystem and is responsible solely for overlays)

11/11/25 - stopped checking controller GUID, realised it was manufacturer id and anyone using generic controllers (like me) would have their controllers lumped onto the same id

11/11/25 - added a debug overlay that shows activity of all input devices

11/11/25 - when refining before final merge, i noticed that SDL2 actually has a limitation with generic controllers, that being it has literally no idea which is which, this isnt a problem if they dont disconnect but finding the right controller to connect back to is now proving extremely difficult

11/11/25 - my first idea for solving this problem is to just reconnect controllers to the slot that doesnt have a valid controller connected, this would work perfectly fine however a foreseeable issue is if two controllers disconnect at the same time then they may end up in the wrong slot. this is the most practical solution for now however once the project is at a later stage my definitive solution is to have a prompt pop up for each disconnected user saying "press any key (player)" so the user can manually put it back in the right slot

11/11/25 - implemented fix to controller reconnection, as suspected theres an issue where disconnected 2 controllers can cause the slots to switch

12/11/25 - noticed that the input sensitivity of controllers depended on the frame rate the game was running on when testing on multiple clamped frame rates

12/11/25 - implemented a dynamic delta time value that can easily be retrieved from RenderSystem and scaled input sensitivity with it

13/11/25 - 22/12/25 - time dedicated to multiple assignments for other modules

23/11/25 - asked myself what should i be doing for users with different display types or render preferences

24/11/25 - utilized nlohmann json to make a generic ConfigManager usable for any config elsewhere in my codebase

25/11/25 - implemented the new config system for video settings, including render resolution, the default render resolution is 1920x1080 since its the most common resolution

27/11/25 - researched ECS implementations for use cases similar to mine

28/11/25 - researched the ogre docs to see what information i should be storing for my ECS entities

29/11/25 - initial ECS implementation with a test cube (different coloured faces) for the demo

12-20/12/25 - final refinements to interim report (extension used)

— out of term development

12/12/25 - research similar split overlay implementations (having overlays unique to a viewport)

14/12/25 - implemented extended functionality to rendersystem through an owned class (PreViewportUpdateListener)

15/12/25 - implemented unique overlay functionality by hiding other overlays at the precise moment they are about to be rendered, this is done using the PreViewportUpdate event that is dispatched automatically by ogre

27/12/25 - limit the cursor position to the size of a unique overlay, this functionality will be extended in future by preventing the cursor from moving over other viewports

— term 2 start

19/01/26 - revisited <https://liacs.leidenuniv.nl/~plaata1/papers/abbadi-resources/entities/actions/a-generalization-118.pdf> for a better understanding of resource entity action

20/01/26 - wrote failing tests for the transformation matrix class

20-25/01/26 - implementation of a rudimentary cursor implementation, noticed significant stability issues from this implementation

26/01/26 - following testing of the cursor implementation, i decided that sticking to the default OS cursor for mouse was a better option for the mouse but keeping the system in place for controllers

27/01/26 - refactored the dispatching order for events such that events are processed the same frame they are created, reducing the perceived delay from the cursor

28/01/26 - created the entity interaction class, a class that's responsibility is to process all entity interactions and route matrixable interactions accordingly.

29-31/01/26 - edge cases for matrix tests, stability improvements following detailed analysis from performance profiler.

02/02/26 - read <https://stackoverflow.com/questions/26516683/reusing-thread-in-loop-c> to learn about reusing threads

02/02/26 - added BS thread pool as a dependency, implemented into the current thread pool pattern, improved fps and stability remarkably (500-2000fps) and cpu usage of the gameinstance->run method dropped from 12.8 percent of 1 core to 2.7 percent

03/02/26 - noticed another significant performance limiter, updated application->run to follow the updated threading model and increased fps from 2000fps to 6000fps (cpu time dropped from 2.7 percent to 1.3)

9/02/26 - designed an initial game map implementation, attempting to replicate the world map as a single mesh

10/02/26 - after experimenting with the initial game map, i realised a cleaner approach would be to separate the land masses as distinct meshes. this realisation came after investigating the ogre doc, specifically the ray tracing functionality. since ogre ray tracing handles mesh collision checks automatically, having the land masses be separate meshes was a clean way of

identifying land the units could spawn on (land masses) and ones they couldnt (the sea)

11/02/26 - redesigned the world map after taking this into account, with countries also pertruding from the sea for an easier time with ogre raytracing.

12/02/26 - updated world manager to load the new globe, each country is added to the same scenenode to make moving the globe easier if need be. the implemented world map does have a couple rendering errors, that being that some countries are visible through the globes surface (when the camera is facing the other side of the globe)

16/02/26 - investigated how input translator currently propogates input events

17/02/26 - realized that input translator up until this point has mainly been focused on only communicating with the same instance thread it runs in (input events arent sent elsewhere but instead the instance thread reads the states manually)

18/02/26 - designed a new struct for input events sent to the render thread to be identifiable back to the instance thread it was sent from (ActionContext), along with other context such as on a MousePressed event it includes the screen coordinates pressed and whether the button was pressed for the first time or is being held down (instead of a MousePressed event only indicating that that device was pressed and nothing else)

27/02/26 - looked into how to fix the rendering errors present from the globe implementation. read <https://forums.ogre3d.org/viewtopic.php?t=81112>

28/02/26 - since ive been using blender for my 3d meshes, the z order should be managed automatically so i deduced that the materials are the issue

28/02/26 - read <https://forums.ogre3d.org/viewtopic.php?t=57151> to identify the issue

28/02/26 - added a depth pre-pass level to the material scripts for the country material, this fixed the issue

6/03/26 - setup for raytracing functionality, initialised an Ogre:rayscenequery on startup

7/03/26 - using the new functionality in input translator, Player general control now starts a raytrace when the mouse is pressed. the result isnt used yet but has been verified to be correct for future implementation

9/03/26 - revisited older overlay code to try and figure out a clean method to add pressable buttons

10/03/26 - decided on using a callback system for button functionality, buttons would register an event to be queued back to an event queue pointer once the button was clicked allowing OverlayController to handle all of the mouse hovering and mouse pressed checks.

11/03/26 - implemented the new functionality into overlaycontroller and added a new button class to make adding a functional button easier (GenericButton.cpp)

14/03/26 - implemented a basic state machine in preparation for a main menu

15/03/26 - using the new button class (Generic button) i created a main menu button that issues ChangeState event when pressed

15/03/26 - added functionality to rendersystem to hide or show a mesh when needed, allowing me to hide the globe by default then show the globe when the play button is pressed

20/03/26 - using previously unintegrated raytracing functionality, players can now select a unit or city

20/03/26 - added functionality to highlight the selected unit or city by changing its colour to red

21/03/26 - extended selection functionality by only allowing one city or unit to be selected, and pressing on a valid land mass will store the coordinates for future movement

22/03/26 - added movement functionality to units, using the stored coordinates then the interaction wheel to move them

25/03/26 - enemy units now appear as a different colour, this was done using a glsl fragment shader that does the ownership check on the gpu. units and cities now use a new material. unit fp that includes this check and once their initialised they have their ownership stored within it

26/03/26 - added a new entity component, ProducesUnitsComponent. in the entitiy loop, this component adds to the available units passively

30/03/26 - added attacking units functionality, final refinements to code

6/04/26 - cpplint compliance 100 percent